

IMO 2026 — Problem 3

Solution by Claude Fable 5

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Source: `results/imo-2026/claude-fable-5/problem-03/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let n be a positive integer. Liu Bang and Xiang Yu have a stick of length 1 and want to divide it between themselves. Liu Bang marks at most n points on the stick, and then Xiang Yu marks at most n points on the stick. The marked points are distinct. Then, the stick is cut at all marked points, creating a number of pieces. Afterwards, they take turns claiming any unclaimed piece of the stick, with Liu Bang going first. Each player's goal is to maximise the total length of their own pieces. For each n , determine the largest value c such that Liu Bang may guarantee a total length of at least c , regardless of Xiang Yu's play.

Solution document

Status

solved

Approaches tried

- pairing-exchange-normal-form — worked (round 1). Lower bound: LB plays the geometric config (pieces $2^{0..2n}$ in units of $1/(2^{\{n+1\}}-1)$); a graph sign-assignment argument on pair/singleton partitions forces $A(M) \geq 1$ unit. Upper bound: XY pairs nearly-equal subset sums (pigeonhole over $2^{\{n+1\}}$ subsets) and realizes the gap as a single leftover via the Folding Lemma with $\leq n$ marks. Reviewer verified all lemmas by hand and both bounds numerically (min-oddsun search $n \leq 3$ exact; strategy simulation $n \leq 5$ including prior counterexamples). APPROVED.
- geometric-cascade-induction — not built this round (Merge Lemma needed repair).
- parity-alternating-sum — not built this round.
- (superseded within pairing slug) naive greedy pairing upper bound — refuted by explicit counterexamples at $n=3,4$; sigma-only induction for the Leftover Lemma cannot close.

Current best

Complete proof of $c(n) = 2^{n/2} \{n+1\} - 1$, both bounds. Solved.

Full proof

Problem. Liu Bang (LB) marks at most n points on a stick of length 1; then Xiang Yu (XY) marks at most n further points, all marked points distinct. The stick is cut at all marked points. The players alternately claim unclaimed pieces, LB first, each maximizing the total length of his own pieces. Determine the largest $c = c(n)$ that LB can guarantee.

Answer. $c(n) = 2^{n/2} \{n+1\} - 1$.

Throughout, write $D = 2^{\lceil n+1 \rceil} - 1$.

0. Setup, conventions, notation

A *multiset* $M = \{a_1, a_2, \dots, a_N\}$ of nonnegative reals is always listed in decreasing order $a_1 \geq a_2 \geq \dots \geq a_N$. Define

- $(M) = a_1 + a_2 + \dots + a_N$ (the total),
- $\text{odds}(M) = a_1 + a_3 + a_5 + \dots$ (the sum over odd ranks),
- $A(M) = a_1 - a_2 + a_3 - a_4 + \dots$ (the alternating sum).

These depend only on the multiset (ties in the sorting do not change the value sequence). Clearly $\text{odds}(M) = ((M) + A(M)) / 2$. (*)

Adjoining elements equal to 0 to M changes none of (M) , $\text{odds}(M)$, $A(M)$: zeros occupy the lowest ranks and contribute 0.

Convention on endpoint marks. A mark at 0 or 1 either produces no piece or a piece of length 0; by the previous paragraph, pieces of length 0 affect nothing in the claiming phase (they have value 0 and sit at the bottom of the sorted order), so we may and do ignore them: all pieces considered have positive length, and each marked point in the open interval (0,1) increases the number of (positive-length) pieces by exactly 1, while marks at 0 or 1 do not increase it.

1. The claiming phase: Lemma 0

Lemma 0 (value of the alternating-claiming game). Let M be a finite multiset of nonnegative reals (the piece lengths). In the game where two players alternately claim elements of M , the mover claiming first, and each player's payoff is the sum of his claimed elements, the first mover can guarantee himself at least $\text{odds}(M)$, and the second player can guarantee that the first mover gets at most $\text{odds}(M)$.

Proof. Define $V(\emptyset) = 0$ and, recursively for nonempty M ,

$$V(M) = \max_{x \in M} [x + (M - x) - V(M - x)].$$

We prove by induction on $|M|$ the two guarantees:

- (i) the first mover has a strategy guaranteeing himself $V(M)$;
- (ii) the second player has a strategy guaranteeing that the first mover gets $V(M)$.

Base $|M| = 0$: trivial. Induction step: For (i), the mover claims an x attaining the maximum; in the remaining game on $M \setminus x$ the opponent moves first, and by (ii) for $M \setminus x$ (roles swapped) the original mover, now second, can hold the opponent to $V(M \setminus x)$; since every element of $M \setminus x$ is claimed by someone, the original mover then collects $(M \setminus x) - V(M \setminus x)$, for a total $x + (M \setminus x) - V(M \setminus x) = V(M)$. For (ii): whatever x the first mover claims, by (i) applied to $M \setminus x$ the second player (who moves first there) can guarantee himself $V(M \setminus x)$, hence the first mover gets $x + (M \setminus x) - V(M \setminus x) = V(M)$. This proves (i) and (ii).

It remains to show $V(M) = \text{oddsun}(M)$. Induction on $|M|$; base clear. Let $M = \{a_1 \dots a_N\}$. By the induction hypothesis $V(M \setminus x) = \text{oddsun}(M \setminus x)$, so the payoff of claiming x is

$$P(x) = x + (M \setminus x) - \text{oddsun}(M \setminus x) = x + \text{evensun}(M \setminus x),$$

where evensun is the sum over even ranks. For $x = a_i$, deleting a_i shifts ranks $j > i$ down by one, so

$$\text{evensun}(M \setminus a_i) = \sum_{\{j < i, j \text{ even}\}} a_j + \sum_{\{j > i, j \text{ odd}\}} a_j.$$

For $i = 1$ this gives $P(a_1) = a_1 + a_3 + a_5 + \dots = \text{oddsun}(M)$. For $i \geq 2$:

$$\text{oddsun}(M) - P(a_i) = \sum_{\{j \text{ odd}, j \geq i\}} a_j - a_i - \sum_{\{j \text{ even}, j < i\}} a_j.$$

If i is odd, this equals $(a_1 - a_2) + (a_3 - a_4) + \dots + (a_{i-2} - a_{i-1}) \geq 0$; if i is even, it equals $(a_1 - a_2) + \dots + (a_{i-1} - a_i) \geq 0$, since the sequence is decreasing. Hence the maximum is attained at $x = a_1$ and $V(M) = \text{oddsun}(M)$.

So after the cutting is done and produces the multiset M of piece lengths ($|M| = 1$), the claiming phase is worth exactly $\text{oddsun}(M) = (1 + A(M))/2$ to LB: he can guarantee this much and cannot be sure of more. The problem is therefore equivalent to:

$$c(n) = \max_{\text{LB cuts}} \min_{\text{XY cuts}} \text{oddsun}(M) = (1 + \max \min A(M)) / 2.$$

We prove: $\max \min A(M) = 1/D$, i.e. LB can force $A(M) \geq 1/D$ (Section 4), and XY can always force $A(M) \leq 1/D$ (Section 5). By (*), this gives $c(n) = (1 + 1/D)/2 = ((D+1)/D)/2 = 2^{n+1}/(2D) = 2^n/D$, as claimed.

2. Elementary facts about alternating sums

Lemma 1. Let $b_1 \geq b_2 \geq \dots \geq b_c \geq 0$ and $U = b_1 - b_2 + b_3 - \dots + (-1)^{c+1} b_c$. Then:

(F1) $0 \leq U \leq b_1$. (F2) If c is odd, then $U \leq b_c$. (F3) If c is even and $c \geq 2$, then $U \leq b_1 - b_c$.

Proof. (F1): grouping from the left, $U = (b_1 - b_2) + (b_3 - b_4) + \dots$ (plus a final $+b_c$ if c is odd), a sum of nonnegative terms, so $U \geq 0$. Also $U = b_1 - (b_2 - b_3) - (b_4 - b_5) - \dots$ (with a final $-b_c$ if c is even), i.e. b_1 minus nonnegative terms, so $U \leq b_1$. (F2): for odd c , $U = (b_1 - b_2) + \dots + (b_{c-2} - b_{c-1}) + b_c \leq b_c$. (F3): for even c , $U = b_1 - (b_2 - b_3) - (b_4 - b_5) - \dots - (b_{c-2} - b_{c-1}) - b_c \leq b_1 - b_c$.

Lemma 2 (peeling a singleton). For any multiset M of nonnegative reals and any $x \in M$:

$$A(M) = A(M \setminus \{x\}) + x.$$

Proof. Say $x = a_i$ in the sorted order of $M = \{a_1 \dots a_N\}$. Removing a_i leaves ranks $j < i$ unchanged and shifts ranks $j > i$ down by one (flipping their signs), so

$$\Delta := A(M) - A(M \setminus \{x\}) = (-1)^{i+1} x + 2 \sum_{j>i} (-1)^{j+1} a_j = (-1)^{i+1} x + 2(-1)^i T,$$

where $T = \sum_{s=1}^{\infty} (-1)^{s+1} a_{i+s}$ is an alternating sum of the decreasing tail, so $0 \leq T \leq a_{i+1}$ by (F1). If i is odd: $\Delta = x - 2T \leq x$. If i is even: $\Delta = -x + 2T \geq -x + 2a_{i+1} - x = 2a_i - x$.

Lemma 3 (peeling a pair). For any multiset M of nonnegative reals and any two elements x, y of M :

$$A(M) \leq A(M \setminus \{x, y\}) + (x - y).$$

Proof. Choose ranks $i < j$ with $a_i = x, a_j = y$ (possible since $x \leq y$; if $x = y$ take any two occurrences). Removing both flips the signs exactly of the ranks strictly between i and j (ranks past j shift by two, keeping their signs), so with $c = j - i - 1$ and $U = \sum_{s=1}^c (-1)^{s+1} a_{i+s}$ (an alternating sum of the decreasing block between them, to which Lemma 1 applies),

$$\Delta := A(M) - A(M \setminus \{x, y\}) = (-1)^{i+1} x + (-1)^{j+1} y + 2(-1)^i U.$$

Four parity cases (note $c \equiv j - i - 1 \pmod{2}$):

1. i odd, j even (c even): $\Delta = x - y - 2U \leq x - y$, since $U \geq 0$ (F1).
2. i odd, j odd (c odd): $\Delta = x + y - 2U \leq x + y - 2a_{j-1} \leq x + y - 2y = x - y$, using (F2) ($U \geq a_{j-1} - a_j = y$). (Here $c \geq 1$, so a_{j-1} is inside the block.)
3. i even, j odd (c even): if $c = 0$, $\Delta = -x + y \leq x - y$ since $x \leq y$. If $c \geq 2$, $\Delta = -x + y + 2U \leq -x + y + 2(a_{i+1} - a_{j-1}) \leq -x + y + 2(x - y) = x - y$, using (F3) and $a_{i+1} - a_i = x, a_{j-1} - a_j = y$.
4. i even, j even (c odd): $\Delta = -x - y + 2U \leq -x - y + 2a_{i+1} \leq -x - y + 2x = x - y$, using (F1).

Definition (pairing partitions and cost). A *pairing partition* Π of a multiset M is a partition of M into blocks of size two (“pairs”) and size one (“singletons”). Its *cost* is

$$\text{cost}(\Pi) = \sum_{\{\text{pairs } \{a, b\}\}} |a - b| + \sum_{\{\text{singletons } \{c\}\}} c.$$

Lemma 4 (Partition Lemma). For every multiset M of nonnegative reals and every pairing partition Π of M ,

$$A(M) \leq \text{cost}(\Pi),$$

with equality for the *adjacent pairing* $\Pi = \{a_1, a_2\}, \{a_3, a_4\}, \dots$ (final singleton $\{a_N\}$ if N is odd):

$$A(M) = \text{cost}(\Pi).$$

Proof. Inequality: peel off the blocks of Π one at a time, applying Lemma 3 for each pair and Lemma 2 for each singleton; after all blocks are removed we reach $A(\emptyset) = 0$, and the accumulated bound is exactly $\text{cost}(\Pi)$. (Lemmas 2 and 3 hold for arbitrary multisets, so the peeling order is irrelevant.) Equality for Π : $\text{cost}(\Pi) = (a_1 - a_2) + (a_3 - a_4) + \dots (+ a_N \text{ if } N \text{ odd}) = A(M)$ by definition.

Lemma 4 is the exchange/pairing engine of this approach: the lower bound will bound $\text{cost}(\Pi) = A(M)$ from below by bounding **every** pairing partition’s cost; the upper bound will exhibit **one** cheap pairing partition.

3. LB's configuration

LB places his n marks at the points $(2^1 - 1)/D, (2^2 - 1)/D, \dots, (2^n - 1)/D$. These are n distinct points of $(0,1)$. The resulting pieces have lengths

$1/D, 2/D, 4/D, \dots, 2^n/D$ (piece k spans $[(2^k - 1)/D, (2^{k+1} - 1)/D]$, length $2^k/D$, $k = 0, \dots, n$),

which indeed total $(2^{n+1} - 1)/D = 1$. From now on we measure lengths in *units* of $1/D$; LB's pieces are $2^0, 2^1, \dots, 2^n$ and the whole stick is D units.

4. Lower bound: $A(M) \geq 1$ unit for every XY response

Call LB's $n+1$ pieces *sources*, indexed $0, \dots, n$ with sizes $s_i = 2^i$ (units). XY places $m \leq n$ marks; by the endpoint convention (Section 0) we may assume all of them lie in $(0,1)$ {LB's marks}, i.e. strictly inside sources. The final multiset M consists of $N = (n+1) + m \leq 2n+1$ *fragments*, each contained in exactly one source; the fragments of source i have positive lengths summing to s_i . (If XY places marks at 0 or 1, N is only smaller, which only helps below.)

Theorem A (lower bound). For every XY response, every pairing partition Π of M has $\text{cost}(\Pi) \geq 1$ unit. Consequently $A(M) = \text{cost}(\Pi) \geq 1$ unit $= 1/D$, and by Lemma 0 and (*), LB is guaranteed at least $(1 + 1/D)/2 = 2^n/D$.

Proof. Fix a pairing partition Π of M . Build a multigraph H :

- vertices: the sources $0, 1, \dots, n$;
- edges: one edge $\{i, j\}$ for each pair $\{a, b\}$ of Π with a a fragment of source i and b a fragment of source j (a loop at i if $i = j$).

The number of edges of H equals the number of pairs of Π , which is at most $N/2 \leq (2n+1)/2 = n$.

Call a connected component C of H *good* if there is a map $\sigma : V(C) \rightarrow \{+1, -1\}$ with $\sigma_i \neq \sigma_j$ for every edge $\{i, j\}$ of C (in particular, C has no loops), i.e. C is bipartite as a multigraph; otherwise *bad*.

Claim: a bad component C has at least $|V(C)|$ edges. Indeed, C is connected, so it has at least $|V(C)| - 1$ edges that join distinct vertices (loops do not contribute to connectivity: if C had a loop and only $|V(C)| - 2$ non-loop edges, its vertices could not all be connected). If C has exactly $|V(C)| - 1$ edges, then none is a loop and, counted with multiplicity, they connect $|V(C)|$ vertices, which forces the underlying graph to be a tree with no repeated edges (a multi-edge would leave $|V(C)| - 2$ distinct adjacencies, disconnecting C); a tree is properly 2-colorable, so C would be good. Hence a bad C has $\geq |V(C)|$ edges. (Claim)

If **every** component were bad, summing the claim over components would give

$$\# \text{edges}(H) \geq \sum_C |V(C)| = n + 1 > n = \# \text{edges}(H),$$

a contradiction. So some component K is good; fix a valid $\sigma : V(K) \rightarrow \{\pm 1\}$.

Assign to each fragment f whose source lies in $V(K)$ the sign $\sigma_f = \sigma_{\text{source}(f)}$. Note that the blocks of Π split cleanly along components of H : a pair joins fragments whose sources are adjacent in H , hence in the same component; a singleton's source lies in one component. Let $\text{cost}_C(\Pi)$ denote the cost contribution of the blocks whose fragments have sources in component

C; then $\text{cost}(\Pi) = \sum_C \text{cost}_C(\Pi)$ and every $\text{cost}_C(\Pi) \geq 0$ (it is a sum of absolute differences and nonnegative singleton values).

For the good component K:

- for each pair $\{a, b\}$ with sources in $V(K)$: the sources are adjacent, so $\text{sgn}_a = -\text{sgn}_b$, hence $\text{sgn}_a a + \text{sgn}_b b = \pm(a - b) \leq |a - b|$;
- for each singleton $\{c\}$ with source in $V(K)$: $\text{sgn}_c c \leq c$ (as $c \geq 0$).

Summing over all blocks in K, and noting that the fragments with sources in $V(K)$ are exactly all fragments of the sources in $V(K)$ (each such fragment lies in some block of Π , and its block stays inside K as just observed), we get

$$\text{cost}_K(\Pi) = \sum_{\{f: \text{source}(f) \in V(K)\}} \text{sgn}_f f = \sum_{\{i \in V(K)\}} \text{sgn}_i \cdot (\text{sum of fragments of source } i) = \sum_{\{i \in V(K)\}} \text{sgn}_i s_i =: W.$$

Replacing sgn by $-\text{sgn}$ (also valid) gives $\text{cost}_K(\Pi) = -W$, so $\text{cost}_K(\Pi) = |W|$.

Finally, $W = \sum_{\{i \in V(K)\}} \text{sgn}_i 2^i$ is a ± 1 -signed sum of **distinct** powers of two over the nonempty set $V(K)$. Let $t = \max V(K)$. Then

$$|W| \leq 2^t - \sum_{\{i \in V(K), i < t\}} 2^i \leq 2^t - (2^t - 1) = 1.$$

Hence $\text{cost}(\Pi) = \text{cost}_K(\Pi) \leq 1$ unit. Applying this to the adjacent pairing Π and using Lemma 4 (equality part), $A(M) = \text{cost}(\Pi) \leq 1$ unit $= 1/D$. By (*), $\text{oddsun}(M) = (1 + 1/D)/2 = 2^{n/D}$, and by Lemma 0(i) LB can claim at least this much.

5. Upper bound: XY can force $A(M) \leq 1/D$ against every LB configuration

Now LB's marks are arbitrary: they produce k pieces of positive length $p_1, p_2, \dots, p_k$ with $\sum p_i = 1$ and $k \leq n + 1$ (each of LB's n marks in $(0,1)$ adds at most one piece). We exhibit an XY response (n marks, all distinct from each other and from LB's marks) after which the final multiset M admits a pairing partition of cost $\leq 1/D$; then $A(M) \leq 1/D$ by Lemma 4, and by Lemma 0(ii) XY can hold LB to $\text{oddsun}(M) = (1 + 1/D)/2 = 2^{n/D}$ in the claiming phase.

Two cutting primitives, each using at most one mark; all marks placed by XY will be strictly inside a *current* piece (a piece of the partition of $[0,1]$ generated by all marks chosen so far), hence automatically distinct from all previously placed marks and from LB's marks — XY of course computes all positions in advance and places them simultaneously; the sequential description below merely defines those positions.

- **Halve(p)**: mark the midpoint of piece p , creating two pieces of length $p/2$ (an equal pair). Cost contribution 0. One mark.
- **Fold(a, b)** for two current pieces with $a > b$: mark the point of piece a at distance b from its left endpoint, creating a piece of length b — which we pair with piece b , an exactly equal pair, cost contribution 0 — and a remainder piece of length $a - b$, which stays in play. One mark. If $a = b$, Fold(a,b) simply pairs the two pieces with **no mark at all**.

Lemma 5 (Folding Lemma). Let A and B be disjoint sets of current pieces, all of positive length, with $v := |A| - |B| \geq 0$. Then using at most $|A| + |B| - 1$ marks XY can cut so that all resulting fragments of the pieces of $A \cup B$ are matched into exactly equal pairs, except for at most one leftover fragment of length $\leq v$.

Proof. Run the following process on the working multisets (A, B) , which maintains the invariant $|A| - |B| = v$:

While $A \neq \emptyset$ and $B \neq \emptyset$: take any $a \in A$, $b \in B$ and apply $\text{Fold}(a, b)$ (to the larger of the two; if $a = b$ no mark is used). If $a > b$, the pair (b, b) is banked at cost 0 and the remainder $a - b$ replaces a in A : the new sums are $(|A| - a + (a - b), |B| - b)$, difference unchanged. If $b > a$, symmetrically the remainder $b - a$ replaces b in B ; difference again unchanged. If $a = b$, both are removed; difference unchanged. Each iteration uses ≤ 1 mark and decreases $|A| + |B|$ by at least 1.

The loop ends with $A = \emptyset$ or $B = \emptyset$. If $A = \emptyset$ and $B \neq \emptyset$ we would have $|B| = -v \leq 0$, impossible for a nonempty set of positive lengths; so $B = \emptyset$ at termination, and the remaining set A satisfies $|A| = v$.

Now fold A down: while $|A| \geq 2$, take any two pieces a, a of A and apply $\text{Fold}(a, a)$; each step banks an equal pair, uses ≤ 1 mark, decreases $|A|$ by 1, and does not increase $|A|$ (the sum drops by $2a \geq 0$). Termination leaves at most one piece, of length $|A| = v$.

Mark count: every iteration of either loop removes at least one piece from the working collection and uses at most one mark; we start with $|A| + |B|$ pieces and end with ≤ 1 , and the last surviving piece never needs a mark, so at most $|A| + |B| - 1$ marks are used in total.

Theorem B (upper bound). For every LB configuration, XY has a response with at most n marks after which $A(M) \leq 1/D$. Hence LB cannot guarantee more than $(1 + 1/D)/2 = 2^{n+1}/D$.

Proof. Case 1: $k \leq n$. XY applies Halve to every piece: $k \leq n$ marks, and M consists of k exactly equal pairs. The pairing partition into those pairs has cost 0, so $A(M) \leq 0$, i.e. $A(M) = 0$ ($A \leq 0$ always, by (F1)); LB gets at most $1/2 < 2^{n+1}/D$ (indeed $2 \cdot 2^n = 2^{n+1} > D$).

Case 2: $k = n + 1$. Consider all 2^{n+1} subsets $T \subseteq \{1, \dots, n+1\}$ and their sums $\sum T = \sum_{i \in T} p_i \in [0, 1]$. Sorting these 2^{n+1} numbers, the $2^{n+1} - 1$ consecutive gaps span at most 1, so by the pigeonhole principle (knowledge_base.md, “Pigeonhole / extremal principle”) two **distinct** subsets $T \neq T'$ satisfy $|\sum T - \sum T'| \leq 1/(2^{n+1} - 1) = 1/D$.

Let $A^* = \{p_i : i \in T \setminus T'\}$ and $B^* = \{p_i : i \in T' \setminus T\}$; these use disjoint index sets, $S := (T \setminus T') \cup (T' \setminus T)$, and $|A^*| - |B^*| = \sum T - \sum T'$. Swapping if necessary, assume $v := |A^*| - |B^*| \geq 0$; then $0 \leq v \leq 1/D$.

XY’s marks: apply the Folding Lemma (Lemma 5) to (A, B) : at most $|S| - 1$ marks, producing equal pairs plus at most one leftover fragment $r \leq 1/D$. Then apply Halve to each of the $n + 1 - |S|$ pieces not in S : $n + 1 - |S|$ marks. Total marks $(|S| - 1) + (n + 1 - |S|) = n$. (Fewer marks are used when some Fold meets equal lengths; “at most n ” is legal.)

The final multiset M consists of exactly equal pairs plus at most one leftover fragment of length $r \leq 1/D$. Take the pairing partition Π : all those equal pairs (cost 0 each) and the singleton $\{r\}$ (if present). Then by Lemma 4,

$$A(M) \leq \text{cost}(\Pi) = r \leq 1/D.$$

By (*) and Lemma 0(ii), XY holds LB to $\text{oddsun}(M) = (1 + A(M))/2 \leq (1 + 1/D)/2 = 2^{n+1}/D$.

6. Conclusion and verification

By Theorem A, LB’s geometric configuration guarantees him at least $2^n/D$ against every XY response and every claiming play (Lemma 0(i)); by Theorem B, against every LB configuration

XY can hold LB to at most $2^n/D$ (Lemma 0(ii)). Therefore the largest guaranteeable value is exactly

$$c(n) = 2^n / (2^{n+1} - 1).$$

Verification. - Tightness at LB's configuration: XY can cut LB's piece 2^n (units) into fragments $2^{n-1}, 2^{n-2}, \dots, 2, 1, 1$ using exactly n marks; the final multiset is $\{2^{n-1}, 2^{n-1}, \dots, 2, 2, 1, 1, 1\}$: n equal pairs plus one leftover unit, so $A(M) = 1$ unit exactly (Lemma 4 both directions: 1 by the pair/singleton partition, 1 by Theorem A) and LB gets exactly $(1 + 1/D)/2 = 2^n/D$. Both bounds are attained, confirming consistency. - $n = 1$: $c(1) = 2/3$. LB marks $1/3$, pieces $(1/3, 2/3)$; if XY halves the $2/3$ -piece, the pieces are $(1/3, 1/3, 1/3)$ and LB takes two of three: $2/3$. Formula: $2^{1/(2-1)} = 2/3$. - Sanity of the answer's form: the constant $D = 2^{n+1} - 1$ is exactly the pigeonhole count of nontrivial gaps among the 2^{n+1} subset sums (upper bound), and exactly the total $1 + 2 + \dots + 2^n$ of LB's geometric pieces, whose distinct-signed-sums property ($|\sum \pm 2^i| \geq 1$) drives the lower bound. - Numerical checks performed during construction (checks only, not proof steps): Lemma 0 against brute-force minimax on 300 random games; Lemmas 2–3 on 20 000 random peel instances; the Theorem B strategy on 5 010 configurations (including the outline-reviewer's counterexamples to the naive greedy, (0.4608, 0.2307, 0.1561, 0.1524) and (0.4388, 0.2228, 0.1296, 0.1271, 0.0817), and all geometric configs $n \leq 7$), always with n marks and value $2^n/D$; 200 000-trial randomized minimization of oddsum over XY responses at $n = 2, 3$ bottoming out at exactly 2^n .